Jose Sanchez Gonzalez

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SUMMARY

M.S. Artificial Intelligence graduate building humanoid hardware, simulation, AR/VR interfaces, and AI software. Develops evaluated retrieval systems and vision-guided robotics with Python/C++, PyTorch, and reproducible tests.

EDUCATION

Oregon State University Corvallis, OR
Master of Science in Artificial Intelligence Jun 2026
Bachelor of Science in Computer Science (Applied: Data Science); Mathematics Minor Jun 2025

TECHNICAL SKILLS

Software: Python, C++, PyTorch, NumPy, Linux, Git, pytest
Robotics and Simulation: Isaac Sim, Isaac Lab, MuJoCo, PPO, ROS 2, ONNX
AI and Retrieval: FastAPI, Docker, PostgreSQL, pgvector, BM25, reranking, MCP
AR/VR and Perception: WebXR, Quest 2, Pinocchio IK, SocketCAN, OpenCV, RGB-D

EXPERIENCE

Graduate Teaching Assistant | Oregon State University Sep 2025 – Jun 2026

- Led weekly two-hour studios for 25+ students in Operating Systems and Computational Methods (COMSOL Multiphysics), teaching C++ concurrency, memory management, file systems, debugging, and numerical simulation workflows.
- Coordinated with course staff to diagnose recurring technical problems, designing targeted demonstrations and communicating debugging strategies across laboratory sessions.

Math and Computer Science Tutor | Oregon State University Nov 2022 – Jun 2025

- Mentored 50+ students one-on-one across mathematics, algorithms, and programming, identifying knowledge gaps and coaching structured quantitative problem solving and independent software debugging.

PROJECTS

Berkeley Humanoid Lite: AR/VR Teleoperation | WebXR, Python, SocketCAN

- Built the physical humanoid and calibrated ten arm joints; connected Quest AR/VR tracking to Pink/Pinocchio inverse kinematics through Python/UDP and SocketCAN.
- Integrated gravity feedforward, grip gating, and camera/lidar/IMU panels; measured one wrist reaching 90% of +/-10-degree step commands in 0.23–0.25 s.

Humanoid Simulation and Policy Evaluation | Isaac Lab, MuJoCo, PPO, ONNX

- Built PPO training, ONNX export, and headless MuJoCo evaluation; compared five randomization settings across 450 episodes with matched controls and saved rollouts.
- Transferred learned navigation to a simulated biped: 35/36 goals, three training seeds, the same twelve unseen mazes, zero falls, and oracle pose/goal.

MetaNaviT: Retrieval and AI Software | PostgreSQL, pgvector, BM25, BGE

- Contributed retrieval/data APIs on a six-person team; extended the fork with hybrid retrieval, BGE reranking, and PostgreSQL/pgvector parity checks.
- Measured +0.075 paired nDCG@10 over BM25 on 300 SciFact queries (95% CI 0.053–0.101), with reproducible evidence tables and drift checks.

Vision-Guided Robotic Pruning | Isaac Sim, RGB-D, OpenCV, ROS 2, C++

- Built RGB-D visual servoing for a simulated UR5e with ToF/contact release gates and an independent sequence grader; retained failure telemetry.
- Reproduced 199/199 recorded decision states in ROS 2 replay; C++17 ToF geometry matched Python to 1e-9 m over 232 valid simulated zones.

SPUR: Metric Depth for Robotic Pruning | PyTorch, DINOv2, FastAPI, ONNX

- Recorded 0.0445 ± 0.0057 m trunk-depth RMSE with three stereo pairs across five stored best-validation runs on synthetic orchard data.
- Packaged Docker/FastAPI inference; measured refiner-only p50 at 156 ms on V100 fp16 with precomputed depth, and checked separate ONNX numerical parity.